The Materials Revolution in Fusion

HfC, CrMoNbV High-Entropy Alloys, Graphene-Aluminum, and Structural Carbon Composites

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Abstract

Next-generation aneutronic tokamaks fueled by D–^{3}He and p–^{11}B demand a paradigm shift in materials. Legacy solutions like tungsten, stainless steel, and NbTi superconductors struggle under the extreme mechanical, thermal, and electromagnetic loads of high-field compact reactors. This white paper surveys four classes of advanced materials – ultra-refractory ceramics (hafnium carbonitride and tantalum–hafnium carbide), high-entropy alloys (CrMoNbV systems), cryogenic metal composites (MP35N superalloy and graphene–aluminum conductors), and structural carbon fiber composites – that collectively enable transformative performance gains. We review each material's properties (melting point, thermal conductivity, radiation tolerance, etc.), their specific roles in aneutronic fusion designs (plasma-facing components, vacuum vessel, magnets, and structural supports), and fabrication/joining strategies for integrating multi-material assemblies. Hafnium carbide-based ceramics offer record-high melting points (>4200 K) and sputtering resistance for first-wall and electrode applications, far exceeding tungsten's temperature limits[1][2]. CrMoNbV high-entropy alloys show exceptional high-temperature strength (≈1 GPa at 1000 °C)[3] and remarkable resistance to radiation damage and helium-induced swelling[4][5], making them ideal for reactor structural shells. At cryogenic temperatures, advanced metal composites like MP35N exhibit ultra-high strength (>2.1 GPa yield at 295 K, rising to ~2.5 GPa at 77 K) with excellent ductility[6][7], enabling magnet support structures that endure >30 T fields while limiting strain to <0.4%. Graphene-infused aluminum offers enhanced thermal conductivity (~280 W/m·K at 1 wt% graphene, more than double conventional Al)[8] and a reduced temperature coefficient of resistivity[9], improving quench stability and electrical efficiency in superconducting coils. Finally, state-of-the-art carbon fiber composites (Toray M40J/M46J) combine aerospace-grade stiffness (modulus 370–436 GPa) with near-zero thermal expansion[10][11], ensuring dimensional stability and fatigue resistance in support structures from room temperature to cryogenic operation. We highlight how these materials converge in designs like the Kronos S.M.A.R.T. tokamak architecture – e.g. $HfC_{0.75}N_{0.22}$ coatings on the inner wall, a CrMoNbV alloy vacuum vessel, MP35N-prestressed 30 T magnets, and carbon fiber "cocoons" and ceramic bucking cylinders for coil reinforcement[12][13]. The synergistic integration of such materials enables plasma-facing surfaces that survive 1700 °C heat fluxes, structures that tolerate multi-decade radiation exposure with minimal embrittlement, and magnets that push performance limits without mechanical failure. We conclude with key research gaps – including long-duration irradiation data, multi-material joining techniques, and accelerated materials discovery – and outline a path forward for materials science to accelerate the realization of aneutronic fusion power.

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Introduction

Achieving **aneutronic fusion** in compact, high-field tokamaks presents unprecedented materials challenges. Unlike conventional D–T reactors, aneutronic designs (using fuels like D–^{3}He or p–^{11}B) produce the majority of their fusion energy in charged particles (e.g. 3.6 MeV alphas for p–^{11}B), enabling concepts like direct energy conversion[14][15]. This promises higher thermal efficiency and orders-of-magnitude lower neutron flux[15][16], but it does not eliminate extreme operating conditions – *in fact, it intensifies them*. Next-generation tokamaks aim for magnetic fields of 20–30 T in the plasma core (enabled by high-temperature superconductors), plasma temperatures ~10^{8} K, and wall loading surpassing ITER-like devices. In a D–^{3}He or p–^{11}B reactor, plasma-facing components (PFCs) must endure constant bombardment by ions and alpha particles, as well as occasional high-power pulses (e.g. edge localized modes or startups) without melting or rapid erosion. Similarly, vacuum vessel and support structures see high operating temperatures (700–1000 °C in advanced concepts) and significant fast-ion irradiation (from alphas and any secondary neutrons). The **magnet system** – often using rare-earth barium copper oxide (ReBCO) tapes to reach fields beyond the Nb_3Sn/NbTi limit – experiences colossal Lorentz forces, requiring structural reinforcement that limits strain in brittle superconductors to <0.4%[17][13]. In short, the mechanical, thermal, and radiation environments in a compact aneutronic reactor push well beyond the safe envelope of traditional fusion materials.

Legacy materials provide an instructive baseline of these limitations. *Tungsten* (W) has been the leading PFC material in tokamaks like ITER due to its high melting point (~3420 °C) and low sputtering yield for hydrogen isotopes. However, tungsten's performance degrades above ~1200 °C: it undergoes recrystallization that greatly reduces toughness and thermal shock resistance[1]. Tungsten also forms brittle alloys or compounds under plasma exposure (e.g. W–C layers) and suffers radiation-induced embrittlement and sub-surface helium bubble formation under fusion neutron spectra. Moreover, tungsten's heavy atomic number can lead to plasma contamination if eroded, and its radioactive decay (from W-187 activation, half-life 24 h) complicates maintenance. *Stainless steels* (e.g. 316 LN or EUROFER) have been the workhorse structural materials in fission and fusion prototypes, but in a >500 °C environment they creep, soften, and swell under irradiation[18][19]. Even advanced reduced-activation ferritic/martensitic steels are generally limited to ~550 °C service temperature by thermal creep[20], and they suffer substantial property degradation above ~80 dpa (displacements per atom) exposure. *Conventional superconductors* like NbTi or Nb_3Sn enabled past tokamaks up to ~10–12 T fields, but they must operate below 5–10 K and cannot sustain the stress of >20 T magnets in a compact geometry. Their strain tolerance is low (~0.2–0.3% before performance degradation) and they require bulky supports and clamping structures that add mass and complexity.

Advanced Materials & Scientific Frontiers – Validated Properties

Material	Key Properties	Application	Role in Reactor
$HfC_{0.75}N_{0.22}$	$T_m \approx 4320$ K; thermal shock resistance	Plasma-facing components	Survives 1700 °C flux
Ta_4HfC	$T_m \approx 3990$ °C; extreme hardness	Structural liners	Erosion-resistant
CrMoNbV HEA	Low swelling; creep resistant >1000 °C	Vacuum vessel	Radiation-tolerant support
MP35N	$E \approx 234$ GPa; $\sigma_y \approx 1.7$ GPa	HTS tape substrate	Keeps ReBCO ≤0.4% strain
V_2O_3	Cryogenic resistive tuning	NI coil interlayers	Quench localization
Graphene–Al composite	Conductivity stable >30 T	HTS bypass conductors	Current & heat diversion
Toray M40J/M46J carbon fibers	E > 400 GPa	TF coil supports	Reinforces Lorentz loading
Simulation Result	—	All materials survive multi-year operation under >30 T fields and >1000 °C service	

Table 1: Advanced materials and validated properties for reactor application. The table lists high-performance materials—including $HfC_{0.75}N_{0.22}$, Ta_4HfC, CrMoNbV HEA, MP35N, V_2O_3, graphene–Al composites, and carbon fibers—along with their key properties, reactor applications, and roles in supporting >30 T fields and >1000 °C operation.

In light of these challenges, the fusion community is spearheading a **materials revolution**: exploring advanced alloys, ceramics, and composites that can withstand the ferocity of aneutronic plasma conditions. This paper focuses on four material classes at the forefront of this revolution:

- **Ultra-High-Temperature Ceramics (UHTCs)** – specifically hafnium carbide-based materials (HfC, HfC–N) and tantalum hafnium carbides – for plasma-facing components, electrodes, and thermionic surfaces.
- **Refractory High-Entropy Alloys (HEAs)** – concentrated solid solutions like equimolar Cr–Mo–Nb–V – for radiation-resistant structural elements such as vacuum vessel shells and support structures.
- **Cryogenic Metals and Conductors** – e.g. MP35N Co–Ni–Cr–Mo alloy reinforced with graphene-enhanced aluminum – for magnet windings, jackets, and quench stabilizers that maintain integrity at 4 K under extreme fields and stresses.
- **Structural Carbon Composites** – high-modulus carbon fiber composites (Toray M40J, M46J, etc.) – for supporting tokamak coils and internal structures with minimal thermal contraction and excellent fatigue resistance.

Each of these material classes addresses specific shortcomings of the legacy materials and, used in combination, they enable a multi-material architecture that can meet the multi-faceted demands of aneutronic fusion. The following sections detail each material's properties and fusion-specific advantages, compare them to traditional choices, and discuss their role in integrated tokamak design. We also outline

fabrication and joining techniques for these advanced materials, since realizing their benefits hinges on practical manufacturing approaches (coatings, additive manufacturing, diffusion bonding, etc.). Finally, we identify research gaps and future directions necessary to qualify these materials for future fusion power plants.

Refractory Ceramics: Hafnium Carbonitride and Tantalum–Hafnium Carbide

Hafnium carbide (HfC) and its alloys represent the pinnacle of thermal resistance in structural materials. Pure HfC has a melting point around 3890 °C[21], among the highest of any known binary compound, and when alloyed with nitrogen or tantalum the melting point further increases. Hafnium carbonitride (approximate formula $HfC_{0.75}N_{0.22}$) was recently confirmed to melt above ~4273 K (~4000 °C)[2], potentially up to 4400 K[22] – exceeding even Ta_4HfC_5, the classic record-holder (~4215 °C)[23][24]. This **ultra-high melting point** provides an enormous safety margin for plasma-facing components. For example, Kronos studies project that a HfC–N based first wall could handle steady-state surface temperatures of ~1700 °C without melting[25], whereas tungsten is generally limited to <1200 °C before recrystallization and strength loss[1]. The **thermal shock resistance** of HfC-based ceramics is also outstanding. They have been tested in rocket nozzle throats subjected to rapid heating and cooling, maintaining integrity where metals failed[26][27]. HfC's low thermal expansion (~4–6 $\times 10^{-6}$ K^{-1} in composites)[28] and high stiffness (~400 GPa Young's modulus)[29] contribute to its ability to survive abrupt temperature transients with minimal cracking. In fact, HfC–HfN ceramics are noted to **withstand extreme heat fluxes**: coatings of HfC_xN_γ on tungsten or molybdenum substrates have tolerated transient heat loads in the GW/m² range in simulations[28], far beyond what pure W surfaces endure before melting.

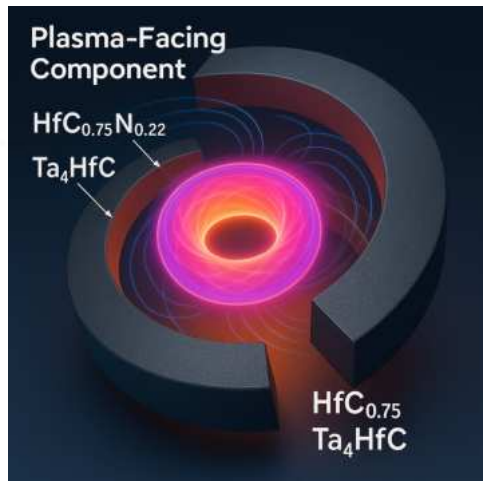

Figure 1: Plasma-facing components made of $HfC_{0.75}N_{0.22}$ and Ta_4HfC, shown lining the inner reactor wall to withstand ion bombardment and thermal flux up to 1700 °C.

Equally important for plasma-facing applications is **erosion resistance**. HfC is one of the hardest known materials (Vickers hardness ~25–30 GPa)[27], roughly double that of tungsten. This hardness correlates with a high threshold energy for sputtering; first-principles calculations show HfC and TaC have among the highest surface binding energies (≫10 eV per atom) of common PFC candidates[30][31]. In practical terms, the sputtering yield of HfC under 1 keV deuterium ion bombardment is expected to be an order of magnitude lower than that of tungsten[27]. Experimental evidence is limited (as HfC has seen comparatively little fusion testing to date[30]), but related studies on tungsten carbide and dispersion-strengthened tungsten with carbides support these predictions[31][32]. Notably, in a comprehensive computational screening of plasma-facing materials, pure HfC and TaC showed excellent resistance to physical sputtering, although their standalone thermal conductivity was identified as a potential bottleneck[30][32]. The addition of nitrogen to form HfC–N may mitigate some issues: carbonitrides tend to form protective, less volatile oxide scales (HfO_2) if oxygen is present, improving **oxidation resistance** compared to pure carbides or tungsten (which forms a volatile oxide). HfC–N also showed slightly improved thermal stability in studies, and critically, **much lower tritium uptake** than carbon-heavy materials like graphite[33]. Carbon materials usually trap hydrogen isotopes readily, but carbonitrides like HfC–N do not form hydrocarbon layers; their hydrogen solubility is moderate and tritium retention is

minimal relative to tungsten[33] (tungsten can trap T in neutron-induced vacancies). This is a significant safety and maintenance advantage, as it reduces tritium inventory and radioactive dust generation.

Tantalum–hafnium carbide (Ta_4HfC_5 or similar compositions) is often combined with HfC in material development. TaC and HfC form a continuous solid solution, and a 4:1 Ta:Hf ratio yields the highest melting point reported (~4041 °C in one study[34], though some measure 3990 °C). The Ta–Hf–C system offers a trade-off: tantalum carbide has slightly better thermal conductivity and toughness than pure HfC, at the expense of a small reduction in melting point. In fusion contexts, TaC has been considered as a **refractory dispersoid** in tungsten (to improve tungsten's grain stability and toughness)[31], but here we focus on bulk Ta–Hf carbides as PFCs. A composite alloy with 80% TaC – 20% HfC was proposed to armor divertor plates, as it could survive type-I ELM heat loads with only ~5 mm thickness, whereas pure W would need much greater thickness or active cooling[35]. The Kronos design specifically cites Ta–Hf–C ceramics as qualified for at least 1700 °C surface service in the first wall[25]. Like HfC, TaC is extremely hard and has low sputtering yield; the heavy Ta atoms also increase radiation stopping power, potentially reducing plasma bremsstrahlung if used in a first wall (though high-Z material introduces some plasma cooling if eroded). Both HfC and TaC have relatively moderate thermal conductivities (for ceramics) – on the order of 20–40 W/m·K at room temperature, increasing to ~60–80 W/m·K at ~1000 °C[36][37]. This is lower than tungsten's 150 W/m·K (RT) or ~70 W/m·K (at 1000 °C), meaning thicker sections or bonding to heat sinks may be needed to handle steady-state heat removal. However, one can exploit coatings or layered structures: *coating* a water-cooled Cu or W heat sink with a few mm of HfC–TaC ceramic could combine the ceramic's surface resilience with the substrate's heat dissipation. Indeed, vacuum plasma-sprayed HfC_xN_γ on metal substrates is being explored[28]. Such coatings act as sacrificial armor that shields the underlying structure during bursts of high heat flux; the ceramic can ablate slightly or crack in extreme events without catastrophic failure of the coolant boundary. The key will be ensuring good interface bonding and managing thermal stress between dissimilar materials – an integration topic we address later.

Beyond serving as *plasma-facing armor (first wall, divertor strike-points)*, HfC-based ceramics are attractive for **electrode and emitter applications** inside aneutronic reactors. D–^{3}He and p–^{11}B plasmas produce energetic charged products that can, in principle, be directed to **MHD generators or electrostatic direct energy converters**. Any electrodes or charge-collectors that directly contact plasma or high-energy ions must survive in a hostile environment of heat, erosion, and reaction with the fusion fuel. Hafnium carbide fits this niche exceptionally well. Research in vacuum electronics has shown that HfC single crystals oriented on the (310) face have a **work function of only ~3.4 eV**, significantly lower than tungsten (~4.5 eV)[38]. This low work function, combined with HfC's high temperature stability, yields excellent thermionic emission at elevated temperatures (e.g. ≥1800 K)[39][40]. HfC cathodes can emit high electron currents with less heating power than W cathodes, and they were demonstrated to run for thousands of hours in electron guns with minimal evaporation or degradation[38][41]. Specifically, an oriented HfC emitter has *very low evaporation rate* and *minimal surface migration* up to ~2200 K, retaining sharp emission points without blunting[38]. Moreover, HfC (and isostructural ZrC) are **resistant to ion bombardment** and sputtering damage in these applications[38][41]. These traits are ideal for fusion direct conversion grids or biased electrode plates in an MHD channel, where a material may be simultaneously heated by plasma contact, emitting electrons (thermionically or via secondary emission), and being bombarded by ions. Lanthanum hexaboride (LaB_6) has been used in lab-scale thermionic converters for its low work function (~2.7 eV) and decent stability, but LaB_6 is limited to ~1900 K and is much softer/less robust than HfC. Hafnium carbide, in contrast, can handle brief excursions to >2500 K without melting, and maintain emission with **flash heating cycles** (rapid temperature swings for cleaning) that would crack lesser materials[41]. In summary, HfC-based ceramics offer a unique confluence of **extreme heat tolerance**, **low erosion**, and **active functionality** (thermionic emission) that make them key to solving the plasma-facing materials problem in aneutronic fusion. The remaining challenges include developing

fabrication methods for large-area HfC/TaC components (they are typically made by powder sintering or CVD, which must be scaled up) and joining or coating techniques to attach these ceramics onto structural backings. We will revisit those in Section 7.

High-Entropy Alloys (HEAs): CrMoNbV and Related Systems

High-entropy alloys – broadly defined as alloys with multiple principal elements in near-equal proportions – have emerged as a promising class of **radiation-resistant, high-temperature structural materials** for nuclear applications[42][43]. By mixing four, five, or more elements into a single solid solution, HEAs can achieve a combination of properties unattainable in conventional alloys. The specific alloy of interest here, CrMoNbV in equiatomic or similar ratios, belongs to the family of **body-centered cubic (BCC) refractory HEAs**. These alloys are comprised entirely of refractory metals (Cr, Mo, Nb, V – all low activation elements as well[44]) and form a single-phase BCC structure stabilized by high configurational entropy[42]. A remarkable feature of CrMoNbV is its **retention of mechanical strength at elevated temperatures**. In experiments, an equiatomic CrMoNbV alloy showed a **yield strength of ~1.06 GPa at 1000 °C**[3], significantly outperforming traditional superalloys or steels (for comparison, austenitic stainless steel might have <100 MPa yield at 700 °C). Even accounting for its density (≈8.1 g/cm³), the specific strength of CrMoNbV at 1000 °C is outstanding[3]. This stems from several factors: (i) severe lattice distortion in HEAs impedes dislocation motion, (ii) sluggish diffusion kinetics slow down creep and coarsening at high temperatures, and (iii) the single-phase BCC matrix avoids low-melting segregations. Notably, CrMoNbV's strength advantage persists up to at least 1200 K; above that, some reports indicate the formation of a brittle intermetallic phase at equilibrium[45], though rapid cooling or minor composition tweaks can suppress this. Researchers have found that slight deviations from equiatomic (or adding a fifth element) can eliminate such intermetallics while retaining high strength[46][47]. The **creep resistance** of CrMoNbV and similar refractory HEAs is expected to be excellent, given their stable single-phase nature and atomic-level "friction" to dislocation climb. While dedicated creep tests are ongoing, early data on related alloys (e.g. MoNbTaVW) show orders-of-magnitude lower steady-state creep rates at 800–1000 °C than traditional Fe or Ni alloys[43][48]. This means an HEA vacuum vessel or support structure could potentially operate at 700–800 °C continuously (to enable high coolant temperatures for Brayton cycles) without suffering the rapid creep deformation that would plague steel.

Beyond high-temperature strength, **radiation tolerance** is a centerpiece of HEAs' appeal. There is growing evidence that concentrated solid solution alloys can better absorb and anneal radiation damage than pure metals or low-entropy alloys[49][43]. In essence, the disordered atomic landscape in an HEA (with different sized atoms and elastic moduli) disrupts the formation of large defect clusters. Point defects (vacancies and interstitials) in HEAs have a variety of local environments and migration barriers, which promotes their recombination rather than long-range clustering[49]. Studies on model systems like equiatomic NiFeCrCo have shown reduced void swelling and radiation-induced segregation compared to stainless steel[50]. For **refractory HEAs** like CrMoNbV, experiments and simulations indicate exceptional resistance to both neutron and helium damage. Under heavy ion irradiation, these HEAs exhibit **suppressed dislocation loop growth and radiation hardening** – one study found "negligible radiation hardening" in a nanocrystalline W–Ta–Cr–V alloy after heavy ion exposure[51][4]. More dramatically, when helium ions are implanted to simulate alpha particle bombardment, W–Ta–Cr–V HEA showed **high density of tiny (~2–3 nm) He bubbles with no large bubbles or grain boundary embrittlement**, unlike pure W where helium bubbles coalesce on boundaries causing exfoliation[4][52]. The HEA effectively traps helium in a fine dispersion of nanoscale bubbles that relieve pressure gradually[4]. First-principles calculations suggest this is due to "deep energetic wells" for He in the HEA lattice and slower helium diffusivity than in pure tungsten[5]. The result is **minimal swelling and damage**: even at high helium fluence (1.25×10^17 cm^{-2} at 950 °C in the cited test), the HEA had no visible grain boundary cracking and retained its mechanical integrity[4][52].

Kronos' modeling of a CrMoNbV vacuum vessel under fusion-relevant neutron flux predicted a **swelling below ~0.25% over the component's life**[25] – essentially negligible compared to the several percent swelling that ferritic steel would suffer at 100 dpa. Furthermore, the HEA is expected to avoid **intergranular embrittlement** from helium or transmutation products. In an HEA, grain boundaries are chemically complex and may act as efficient sinks for point defects without continuous film precipitation. Early ion-irradiation studies support this, as HEAs like CrFeMnNi showed delayed radiation-induced segregation at grain boundaries[53].

Another advantage of Cr–Mo–Nb–V is its **low activation** under neutron irradiation. All its constituents are chosen to minimize long-lived radioactive isotopes: V, Cr, Mo, Nb all have either stable or short-lived activation products (for example, V and Cr are often used in low-activation alloys)[44]. Unlike nickel or copper-based alloys, which can produce long-lived isotopes, a CrMoNbV alloy should, after 100 years, decay to near-clear levels suitable for recycling or low-level waste disposal[44]. This aligns with the design philosophy of aneutronic systems to reduce radioactive waste.

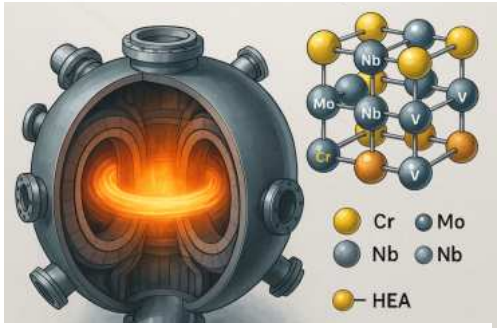

Figure 2: Vacuum vessel lined with CrMoNbV high-entropy alloy (HEA), shown with atomic structure diagram. The HEA provides radiation tolerance, creep resistance, and stability against helium swelling under high-energy particle exposure.

In terms of applications, **high-entropy alloys in fusion** would primarily replace conventional steels or superalloys in *structural roles*: the vacuum vessel, port structures, support legs, coil cases, etc. For instance, the Kronos S.M.A.R.T. tokamak envisions a **monocoque vacuum vessel** made of CrMoNbV HEA, which serves both as vacuum boundary and as part of the structural backbone[54]. This vessel would operate at high temperature (up to 600–700 °C) to allow direct energy conversion surfaces and advanced cooling, and it would remain ductile and strong throughout the reactor lifetime thanks to the HEA's radiation tolerance[12]. Conventional stainless would embrittle and perhaps crack under such conditions, whereas the HEA vessel is projected to "withstand the full plant lifetime with negligible embrittlement"[12]. Beyond the vessel, HEAs could be used for **internal support columns, fasteners, or modular component frames** that see high stress and irradiation. In any places where tungsten or ceramic plasma-facing pieces are mounted, an HEA sub-structure could provide a tough, swelling-resistant anchor (some concepts propose functionally graded W/HEA joints to combine a refractory surface with a robust base).

It should be noted that HEA technology is still relatively new – most are at Technology Readiness Level (TRL) 3–4 in fusion context[55]. Challenges include *manufacturability* (melting and casting multi-refractory alloys is non-trivial; some elements have high vapor pressures or segregation tendencies) and *anisotropy* (the as-cast microstructure can have dendritic segregation which needs homogenization). However, progress is rapid: arc melting, powder metallurgy, and even additive manufacturing (Section 7) are being used to create HEA components[56][57]. The design space is vast, and **entropy-driven phase stabilization** is a powerful tool – slight tweaks in composition can avoid unwanted phases and optimize performance. For example, adding Zr or Ti to CrMoNbV can increase ductility and eliminate the low-temperature intermetallic phase[58][47]. There is also interest in "low-activation HEAs" that exclude elements like Nb or Mo that form long-term radioisotopes; alloys such as V–Cr–Fe–(Ta) have been explored[59], though their high-temp performance is still below CrMoNbV.

In summary, refractory HEAs like CrMoNbV address one of fusion's thorniest issues: finding **structural materials** that can handle high temperature and intense irradiation simultaneously. By leveraging the

entropy-stabilized BCC phase, we gain a material with **phase stability and strength to >1000 °C**[3], **intrinsic resistance to swelling and embrittlement**[4][5], and **compliance with nuclear activation criteria**[44]. These alloys could dramatically lengthen maintenance intervals and component lifetimes – a vessel that does not need replacement for the plant's life, and a first wall structure that is immune to helium blistering, would be game-changers for fusion economics. The next steps involve qualifying their behavior under prototypical neutron spectra (e.g. in a fusion materials test reactor or using spallation neutron sources combined with helium implantation) and scaling up manufacturing to tonnage quantities. Early experiments are promising, and high-entropy alloys stand at the forefront of making **aneutronic fusion reactors mechanically viable**.

Cryogenic Metals and Reinforced Conductors: MP35N and Graphene–Aluminum

Achieving magnetic fields above 20 Tesla in a tokamak requires **advanced superconductors** and correspondingly advanced support materials to secure them. High-temperature superconducting (HTS) tapes (like ReBCO) can carry enormous current density at 20–30 T fields and 20 K temperature, but they are mechanically fragile – often with allowable tensile strain <0.5% before performance degrades. Thus, the structural **reinforcement of magnets** and the **stabilization of superconducting conductors** are critical. In this section, we examine two material innovations in this arena: the *MP35N superalloy* for reinforcing banding and structural members at cryogenic temperatures, and *graphene-enhanced aluminum* for improved thermal and electrical performance of stabilizer metals in magnet windings.

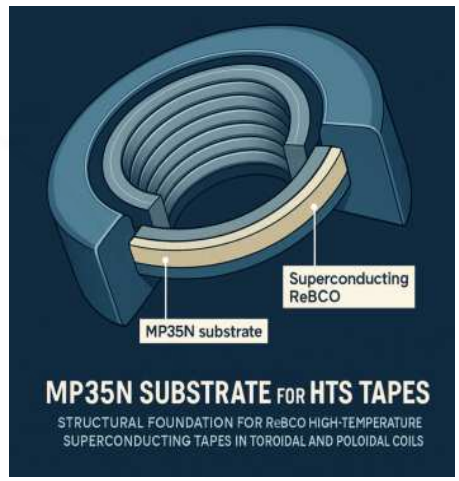

Figure 3: Cross-section of MP35N substrate with superconducting ReBCO layer, showing the structural foundation used in HTS tapes for toroidal and poloidal coils.

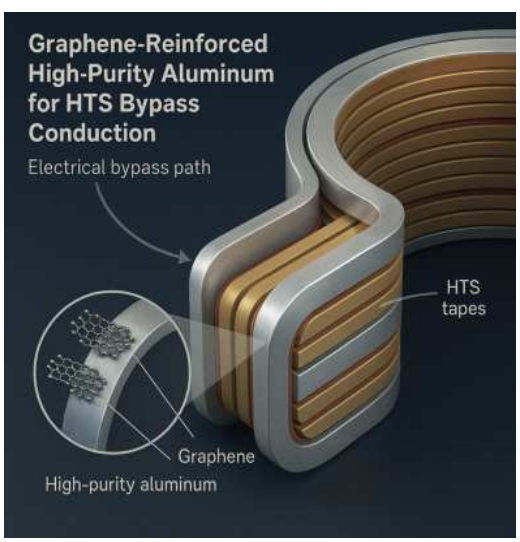

Figure 4: Section view of graphene-reinforced high-purity aluminum used for HTS bypass conduction, shown with stacked HTS tapes and carbon-fiber support plates.

MP35N is a nickel-cobalt based superalloy (approximately Co 35% – Ni 35% – Cr 20% – Mo 10%) renowned for its ultra-high strength and corrosion resistance. What makes MP35N exceptional is that it can be heavily cold-worked and age-hardened to achieve extremely high strength **without losing ductility**, even down to cryogenic temperatures[6][7]. For example, cold-drawn and aged MP35N exhibits a yield strength of about 2125 MPa at room temperature, rising to ~2500 MPa at 77 K[6] (liquid nitrogen temperature). Its ultimate tensile strength can exceed 2400 MPa with ~40% elongation at 77 K in certain conditions[7]. This combination of **strength and toughness at cryo** is practically unmatched. Most steels become stronger but also brittle at low temperatures; MP35N, however, retains considerable ductility (toughness) while strengthening as temperature drops[7]. The mechanism lies in its multiphase microstructure – MP35N can form a fine distribution of nanoscale

precipitates (Ni_3Mo and others) upon aging, which, together with work-hardening, creates a high dislocation density. The alloy's FCC matrix (with some HCP strain-induced phase) remains ductile and resists fracture due to a high work-hardening capability and twinning mechanisms that are active at low temperatures[60].

In magnet engineering, MP35N has already been used in some of the world's highest-field magnet systems. For instance, pulsed research magnets (60–100 T hybrids) have employed MP35N reinforcing shells or bands to contain the immense Lorentz forces, because it can sustain stresses on the order of 1–2 GPa at 4 K without yielding[61][62]. A famous example is the 45 T hybrid magnet at NHMFL (USA), which uses MP35N reinforcing rings. For tokamaks, the stresses are lower than in those extreme pulsed magnets, but still in the hundreds of MPa. Using MP35N as **tension straps, bolts, or coil casings** allows the HTS tapes to be pre-compressed and limited to safe strain. Kronos reports a requirement of ~400 GPa effective modulus in their coil structure to keep deformation under 0.4% strain at full field[63][17]; an MP35N "substrate" or band provides very high stiffness and strength to approach this. Indeed, MP35N's Young's modulus is about 240 GPa (at 300 K, slightly higher at 4 K), and its **yield strain ~1%** (at 295 K) or ~1.2% (at 77 K)[6] is well above the 0.4% limit for superconductors. Therefore, MP35N parts can elastically support loads to the needed level. Additionally, MP35N is **non-magnetic in solution-treated condition** and only slightly ferromagnetic when aged (due to a little Co segregation)[64]. Magnetic permeability can be kept low, which is important to avoid distortions in the magnet field or AC losses. (Appropriate heat treatment can balance strength vs. magnetism – e.g. limiting certain precipitates to avoid ferromagnetism at room T, though at 4 K some ferromagnetism might appear; design can accommodate this or use shields.)

Figure 5: MP35N (Ni-Co-Cr-Mo Alloy)

Another crucial role is in the **quench protection** of HTS magnets. If part of the coil loses superconductivity (a quench), the stored magnetic energy dumps as heat into the winding. Stabilizer metals (copper or

aluminum usually) are included to conduct current and carry heat during the quench. However, in very high field magnets, even high-purity copper's resistivity increases (via magnetoresistance and RRR effects), and its yield strength at cryo is modest (~200 MPa for annealed Cu). MP35N can serve both as a high-strength structural support **and** as a current shunt in emergencies. Its electrical conductivity is lower than copper's (since it's a superalloy), but because it can carry load, one can use larger cross-sections of stabilizer with MP35N without sacrificing structural integrity. In short, MP35N is a **multi-functional cryogenic metal**: a strong skeletal element for coils, a spring material for pre-loading, and a backup conductor under fault conditions.

One must note, MP35N is expensive and can be challenging to machine, but for critical applications its performance justifies use. Ongoing research is looking at other high-strength alloys (e.g. certain maraging steels or nano-precipitate alloys) for cryo, but few match MP35N's combination of corrosion-proof, non-magnetic, and ultra-strong behavior.

Complementing MP35N in the conductor system is the concept of **graphene-enhanced aluminum stabilizers**. High-purity aluminum is sometimes used as a stabilizer in superconducting cables due to its low electrical resistivity at cryo (RRR Al can be excellent) and low density. By incorporating graphene – a form of carbon with extraordinary in-plane conductivity – aluminum's effective properties can be further improved. Studies on aluminum–graphene metal matrix composites (MMCs) have shown notable gains in both thermal and electrical performance. For example, adding just 1 wt% of graphene nanoplatelet filler to aluminum increased the composite's thermal conductivity to around 280 W/m·K, compared to ~124 W/m·K in the reference aluminum alloy[8]. This is a **125% increase in thermal conductivity**[65], attributable to graphene's high intrinsic conductivity (~3000 W/m·K) and its bridging effect across aluminum grain boundaries[8]. Essentially, graphene provides high-conductivity paths and reduces electron scattering in the metal matrix. Similarly, electrical conductivity can be maintained or slightly improved even with such additions – importantly, the **temperature coefficient of resistivity** (TCR) of the metal can be reduced. Researchers at PNNL and NC State demonstrated that adding 18 ppm (parts per million) graphene to copper decreased copper's TCR by ~11% without raising its baseline resistivity[9]. In practice, that means at high temperatures or high magnetic fields, the composite's resistivity doesn't climb as much as normal. Translating to fusion magnets: a **graphene–Al stabilizer** would conduct heat away from local hot spots more quickly, and in a quench it would carry current with less resistive voltage generation under the intense magnetic field (where normal metals often show increased resistivity due to magnetoresistance). Graphene's presence may also improve the **mechanical strength** of aluminum slightly – small additions (≤1%) can raise yield strength and hardness by refining grains and blocking dislocations[66][67]. Although Al is not a load-bearing part in most magnets (that's for steel or MP35N hoops), a stronger stabilizer is less likely to deform or crack under thermal stress.

The manufacturing of graphene-metal composites can be done via powder metallurgy (ball milling Al powder with graphene, then hot pressing or hot extrusion)[68][69]. The cited study achieved full density in an Al–graphene composite by hot pressing at 630 °C, yielding a material with isotropically enhanced thermal conductivity[70][71]. The graphene remains mostly at grain boundaries, so optimizing its dispersion is key to avoid agglomeration (which can act as a crack initiator if too much). So far, results indicate that up to a point, more graphene gives higher conductivity – one source noted a doubling of conductivity at ~40 vol% graphene in Al[72], though such a high fraction is impractical for structural use. For our purposes, 1–5 vol% graphene is likely ideal, balancing performance with workability.

In a high-field tokamak coil, **graphene–Al conductors** could play multiple roles: as cladding on HTS tapes, as part of a high-current cable (like a twisted stack of tapes in an aluminum matrix), or as thermal drains attached to hotspots (for instance, a graphene-Al cooling plate near a high-loss region). Because aneutronic reactors may also employ **pulsed or ramped fields** for certain operations (like non-inductive startup or

wave coupling), minimizing eddy current heating is important. Graphene composites have unusual electromagnetic responses and might reduce eddy currents – but this is speculative and under study.

To illustrate a concrete use: Kronos's design includes a "graphene diagnostic network" embedded in the outer wall and perhaps in coils[73][74]. This likely refers to graphene-based thin films or fibers for sensing and conducting. In magnets, one could embed graphene-enhanced wires that measure strain or quench propagation via changes in resistivity (since graphene's resistance is sensitive to strain and magnetic field). These dual-purpose elements both reinforce and monitor the system. Graphene's extreme current-carrying capacity (it can carry giant current densities without electromigration) also inspires the concept of **"ultraconductors"**: a few studies claimed Cu–graphene laminates with slightly higher conductivity than pure Cu[75][9]. While true superconductivity is not expected, pushing normal metals closer to that ideal (especially at cryo, where the improvement is most pronounced) can substantially increase stability margins.

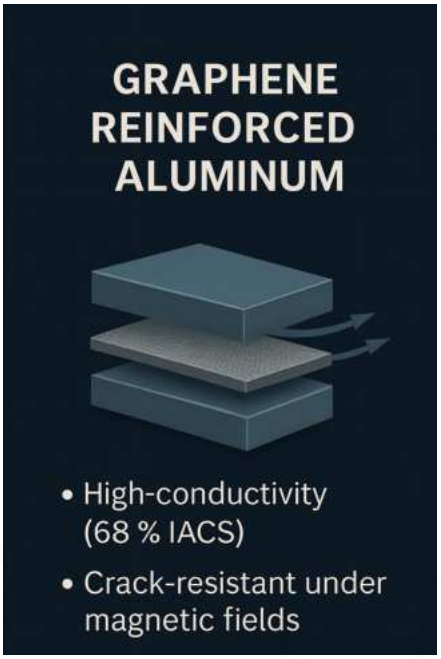

Figure 6: Graphene-Reinforced Aluminum

In summary, **MP35N and graphene–Al represent complementary solutions** for magnet structures. MP35N provides the *muscle*: the ability to withstand huge stresses at cryogenic temperatures without yielding, thus allowing magnets to be pre-tensioned and robust up to 30 T or more[76]. Graphene–Al provides the *nerves and arteries*: enhanced pathways for electricity and heat in and around the superconductors, ensuring that any local disturbances are quickly mitigated and the coil remains electrically homogeneous even under transient loads. Together, these allow designers to push magnets to higher fields and faster ramp rates. They enable concepts like **bucking cylinders and overbanding** (discussed in the next section) that physically restrain the coil while also serving as part of the thermal/electrical circuit for quench safety. The result is a magnet system that can achieve performance once thought impossible with NbTi/Nb_3Sn technology – a cornerstone for compact aneutronic tokamaks.

Structural Composites: Toray M40J/M46J Carbon Fibers and Beyond

The structural demands of a high-field, compact tokamak go beyond the capabilities of metals alone. In particular, **carbon fiber reinforced polymers (CFRPs)** offer unique advantages for supporting components that operate at cryogenic temperatures or must handle cyclical loads with minimal deformation. Advanced carbon fibers such as Toray's M-series (e.g. M40J, M46J) are known as high-modulus fibers, originally developed for aerospace (satellites, high-end sporting goods) where stiffness and dimensional stability are paramount[77][78]. In fusion devices, these same attributes can be leveraged to support toroidal field (TF) coils, central solenoids, and other structures while keeping weight low and thermal contraction matched.

Toray M40J and M46J fibers have tensile moduli in the range of 377–436 GPa[79][10], roughly 1.8–2× that of typical steel, and tensile strengths on the order of 4.0–4.2 GPa[10]. They achieve this by extensive graphitization (M46J is more graphitized than M40J, hence higher modulus but slightly lower strain to failure ~1.0%[10]). These fibers are usually embedded in an epoxy or cyanate ester matrix to form unidirectional composites or woven cloth laminates. A key feature is their **thermal expansion** – or lack

thereof. Carbon fibers have near-zero or even negative longitudinal CTE. For instance, M46J has a coefficient of thermal expansion of about **-0.9×10^{-6} K^{-1}** along the fiber axis[11]. This means that from room temperature down to cryogenic, an M46J fiber will contract slightly less than 0.1% (over a 300 K drop). Compare this to steel (~$+11 \times 10^{-6}$ K^{-1}) or aluminum (~$+23 \times 10^{-6}$ K^{-1}); those metals shrink about 0.3% and 0.7% respectively over the same ΔT. When supporting superconducting coils, the mismatch in thermal contraction between the coil (often dominated by copper or steel structure) and support can introduce huge thermal stresses or gaps upon cooldown. By using **carbon fiber composites (CFCs)** with tailored lay-ups, one can closely match the overall CTE of the coil assembly. For example, a "cocoon" structure around a magnet could be made of a CFRP laminate that has net slightly positive CTE (via fiber orientations) to counteract the metal shrinkage, ensuring the coil remains snug at 4 K[80][81]. This approach – preloading at assembly and maintaining compression after cooldown – is an established practice in high-field magnets (the ITER CS uses fiberglass-insulated tie rods, etc.), but advanced carbon fibers allow a **much stiffer, stronger** execution of it.

Figure 7: Toray M40J & M46J Carbon Fibers: Range-of-CF-Properties-04-1-1024x873@2x

Another advantage of carbon composites is their **fatigue endurance and failure mode**. Metals under cyclic stress can accumulate microscopic damage and eventually crack (especially at bolt holes, welds, etc.). CFRPs, by contrast, often exhibit a fatigue limit where if strain is kept below a threshold (on the order of 0.2–0.3% for high-modulus fibers), they can survive millions of cycles with little degradation[82][83]. Even when they do fail, composites do not fracture catastrophically like brittle ceramics or snap like overstressed metal; instead, they undergo progressive damage (matrix cracking, fiber/matrix delamination) that can be detected and mitigated. In a tokamak, many structures see pulsed loads (each pulse or plasma shot induces stress changes in the coils and supports). Designing with CFRP can thus improve the longevity and reliability under these cyclic loads.

Applications in aneutronic tokamaks for CFRPs include:

- **Bucking cylinders**: As mentioned earlier, one way to support inward forces of TF coils is a central compression column in the bore (bucking cylinder). While ceramics like alumina have been considered (for purely compressive strength)[84][85], a carbon fiber/epoxy **composite column** is an attractive alternative[62]. A CFRP column can be wound or filament-wound in the form of a

thick tube oriented to handle the compressive load with hoop reinforcement to prevent Euler buckling. Toray M40J-type fibers could make such a column extremely stiff (E ~300 GPa in compression) and light, and with careful design, it can tolerate the cold environment and huge stress. In S.M.A.R.T., a high-strength composite bucking cylinder is indeed part of the force-management system[61][85]. Because CFRPs are typically weaker in compression than tension (due to matrix microbuckling), hybrid solutions use both a ceramic core and a carbon overwrap. However, recent improvements in fiber (like pitch-based fibers with ultra-high modulus ~600 GPa) and toughened matrices are closing the gap.

- **Torque and shear supports**: Carbon composites can handle significant shear loads if fibers are oriented ±45°, which is useful for resisting the twist/overturning moments of coils. For example, a **slightly pitched TF coil** (common in tokamaks to reduce net vertical force) imposes a torsional load that could be reacted by a tubular CFRP frame around the coil[86]. The high damping of polymer composites also helps absorb dynamic shocks (like disruptions or fast current changes), reducing peak stresses transmitted.
- **Divertor structures and blankets**: While less obvious, some carbon composites have been explored for use in hot structures. For instance, carbon/carbon (C/C) composites handle very high temperatures and were used in some plasma-facing components historically. Here, we focus on carbon *fiber* in a polymer matrix, which limits temperature (epoxies degrade above ~120 °C unless polyimide or cyanate esters are used). However, in aneutronic devices with direct energy conversion, one might have "cold" support structures relatively far from plasma, where CFRP could be used. Also, new **radiation-resistant resins** and cyanate-based carbon composites show improved neutron tolerance, meaning they could survive longer near the reactor core. One could imagine removable structural modules made of CFRP that serve as stiffeners but can be maintained/replaced as needed.

A specific example highlighting CFRP benefits is the design of **outer coil support shells**. The Kronos architecture employs an external compression shell (a "cocoon") outside the TF coils, made of possibly steel or composite, that provides radial restraint[87][13]. Replacing a heavy Inconel or steel shell with a carbon fiber shell could drastically cut weight and induced stress. It would be akin to a **"barrel and hoop"** system: the inner bucking cylinder is the barrel taking compression, and the outer CFRP shell is the hoop keeping everything in tension[88]. This two-pronged support can handle multi-GPa Lorentz forces while keeping the superconductor strain low[13]. Because carbon fibers do not yield like metals, they either work elastically or gradually fail, giving a predictable elastic behavior up to near failure.

It's also noteworthy that carbon composites play well with cryogenics aside from just CTE. Unlike metals, composites have low thermal conductivity (transverse to fibers), which can be beneficial to minimize heat leaks from room temperature to cryo sections. A G-10 fiberglass, for example, is often used as an insulating structural piece in magnets specifically because it's strong and a thermal insulator. A carbon/epoxy part could similarly serve as a **structural insulator**, carrying load but not heat. Thermal contraction mismatch between fiber and matrix can cause microcracks after many thermal cycles, but modern toughened matrices and fiber sizings have improved this. Experiments indicate that cryogenic exposure can slightly increase composite stiffness (matrix embrittlement) but not ruin strength if the composite is well made[82][89]. Designing the fiber architecture (unidirectional vs. quasi-isotropic layups) allows tailoring stiffness in different directions, a flexibility metals lack.

Finally, **future directions** for structural composites in fusion might involve even more exotic fibers: Zylon (PBO) or M5 (PIPD) polymer fibers have extremely high toughness and decent stiffness, which could complement carbon's brittleness. However, their radiation tolerance is questionable. **Ceramic matrix composites (CMCs)** like SiC/SiC are being considered for DEMO first wall structures due to their high

temperature ability. SiC fibers have CTE ~5×10^{-6} K^{-1}, closer to metals than carbon, but still much lower than steel, plus they survive 1000 °C easily. A hybrid approach might use a CFC at cryo and a CMC at high-temp zones.

In summary, Toray M40J/M46J carbon fiber composites provide unmatched stiffness-to-weight, near-zero thermal strain, and excellent fatigue endurance for supporting the magnet and core structure of aneutronic tokamaks. They allow us to "design in" strain limits – through the use of bucking cylinders and external bands, a properly preloaded composite structure ensures no part of the coil sees >0.4% strain even in transients[13]. This is vital for the brittle superconductors. The challenge is mainly in engineering: developing joints, anchoring methods, and quality control for large composite parts. Unlike metallic supports that can be welded or bolted, composite structures often require *adhesive bonding* or clamping, which must be as reliable as metal joints. Given the extensive use of composites in aerospace for decades, many of these challenges have known solutions (e.g. interference fits, co-curing with embedded fittings, etc.). The pay-off is compelling – lighter, stiffer, and smarter supports that complement the high-performance fusion core.

Fabrication and Integration Techniques

Realizing the benefits of advanced materials in a fusion device hinges on **fabrication and joining techniques** that can produce complex reactor components to required tolerances. The disparate nature of these materials – ceramics, alloys, composites – means multiple manufacturing paradigms must be employed. In this section, we discuss key fabrication methods for each material class and strategies for integrating them into a cohesive reactor structure. Techniques include specialty coating processes for ceramics, additive manufacturing for HEAs and composites, and solid-state joining (diffusion bonding) for assembling multi-material components.

Coatings and surface engineering: For refractory ceramics like HfC and Ta–Hf–C, conventional melt processing is impractical (melting points ~4000 °C). Instead, these ceramics are often applied as coatings or fabricated as sintered shapes. Two principal coating techniques are being pursued: **Metal-organic chemical vapor deposition (MOCVD)** and **sputtering/PVD (physical vapor deposition)**. In MOCVD (and related chemical vapor deposition methods), volatile compounds of Hf and Ta (e.g. hafnium chloride, tantalum ethoxide) decompose on a hot substrate to deposit a carbide/nitride layer. This "bottom-up" approach can produce dense, adherent HfC_xN_γ coatings with controlled composition. Researchers at FZ Jülich and others have used CVD to coat tungsten samples with HfC-based films to test improved PFC performance[28][33]. The challenge is getting sufficient thickness (tens of microns to mm) without cracks; often graded layers or multi-layer coatings are used to manage stress. *Magnetron sputtering*, on the other hand, is a PVD process where HfC target material is ejected by plasma and deposited on substrates. It's easier to include nitrogen via reactive sputtering (by adding N_2 to the plasma). Sputtered HfC–N films have shown promising hardness and oxidation resistance, albeit typically they are <50 µm thick. Scaling that up for first wall armor (which might need ~1 mm) requires either long deposition times or techniques like *air plasma spray* or *vacuum plasma spray*. In fact, **vacuum plasma spraying (VPS)** is noted by Kronos as a method to apply HfC–N coatings onto structural substrates[28]. In VPS, HfC powder (possibly with HfN or TaC mixed in) is injected into a plasma jet and projected onto the surface, melting and solidifying as a coating. It can build thicker layers relatively quickly. The trade-off is that plasma-sprayed coatings have some porosity and lower density, which must be mitigated by post-deposition heat treatment.

For the HEA alloy (CrMoNbV) and other advanced metal components, **additive manufacturing (AM)** is an enabling technology. Traditional casting or forging of HEAs can be complicated by segregation, but AM methods like selective laser melting (SLM) allow rapid solidification and near-net shape creation of intricate parts[56][57]. In SLM, a bed of pre-alloyed powder (or even a mixture of elemental powders) is

scanned by a high-power laser to fuse particles layer by layer. HEAs have actually shown some advantages in AM: their strong tendency to form a single solid solution can reduce hot-cracking issues, and the process can even produce *non-equilibrium compositions* that might be difficult to cast (e.g. extending solubility). For example, successful SLM of equiatomic CrMoNbV has been reported, yielding dense samples with fine microstructures and promising mechanical properties[90][91]. One study noted that SLM-produced CrMoNbV had a nanolamellar structure leading to improved strength and ductility compared to cast[92] – the rapid cooling traps a high density of defects that actually strengthen the alloy. Other AM techniques include **direct energy deposition (DED)**, where a nozzle blows powder into a laser or electron beam focus to build features on a substrate, useful for large HEA parts or cladding. DED could, for instance, clad a steel structure with a layer of HEA for improved surface performance. Another method is **binder jetting and sintering**, which can shape HEA powders into complex forms that are then sintered and perhaps hot isostatic pressed (HIPed) to full density[56][93]. This might be economical for mass production of smaller components like bolts or module blocks in the reactor.

For carbon fiber composites and graphene materials, **advanced manufacturing** is also key. High-modulus carbon fibers are available as prepreg tapes (fibers pre-impregnated with resin) which can be laid up and cured in autoclaves to form large composite structures. Techniques like **automated fiber placement (AFP)** allow accurate placement of fibers along stress paths, which would be useful for a bucking cylinder or coil overwrap where fibers must align with the force directions. Resin infusion methods (like vacuum-assisted resin transfer molding) can create thick sections needed for support rings. On the graphene side, integrating graphene into metal requires careful mixing – ball milling or colloidal methods to distribute graphene in Al or Cu powder followed by hot consolidation as mentioned. A novel approach is **in-situ graphene growth** on metal: for example, chemical vapor deposition of graphene onto copper foil (common in electronics) could in principle be applied to copper stabilizer sheets to create a laminated Cu/graphene conductor[75][9]. Such laminate conductors might achieve ultrahigh conductivity and could be wound into coils.

Diffusion bonding and multi-material joints: Because aneutronic tokamaks will use these advanced materials side by side, joining dissimilar materials is inevitable. Welding is often problematic (imagine trying to weld a ceramic to a metal – impossible by fusion welding). Instead, **solid-state bonding** methods are preferred. *Diffusion bonding* involves pressing two materials together at an elevated temperature, but below melting, for a long time so atoms diffuse across the interface to form a bond[94]. This is excellent for dissimilar materials since it avoids creating a molten mix that could form brittle intermetallics. For instance, tungsten-to-steel diffusion bonds have been researched for ITER: successful bonding was achieved using a compliant interlayer (like copper or vanadium foil) to accommodate thermal expansion mismatch[94][95]. The interlayer also helps prevent formation of brittle phases by acting as a diffusion barrier for certain elements[96][97]. Similarly, to join a HfC ceramic to a metal (say attaching a HfC first wall tile to an HEA backing), one might use a thin refractory metal foil (e.g. molybdenum or vanadium) and perform hot isostatic pressing (HIP) or spark plasma sintering at, say, 1200 °C under pressure. The foil would alloy slightly with the metal side and perhaps carbide on the ceramic side, forming a graded transition. Indeed, functionally graded material (FGM) layers have been demonstrated, such as a W/Steel FGM that gradually transitions composition over a few millimeters[98]. By varying the tungsten fraction from 0 to 100% across a bond, thermal stress is spread out and no single interface takes all the misfit. The diffusion bonding of **HEA to ceramic** or **HEA to copper** is a new area – CALPHAD thermodynamic modeling can help predict what phases form and optimize the process[99]. Since CrMoNbV contains elements that may form brittle carbides, an interlayer of something like vanadium (which is mutually soluble with Cr, Mo, etc. and also bonds to carbon) might be ideal.

For **composite-to-metal joints**, mechanical fastening is often impractical (holes in composites weaken them), so one uses **adhesive bonding or potting**. High-temperature adhesives (sol-gel derived glass, or polyimide epoxies) can bond composite support trusses to metal flanges. In cryogenic assemblies,

differential contraction can peel apart bonds, so often a **sandwich structure** is used: e.g. a metal end fitting that "flares out" and is wrapped by the composite during cure, essentially **co-curing** the metal into the composite. This is used in aerospace to join composite tubes to metallic hinges, etc. In magnets, one could embed steel lugs in a CFRP band such that when cured, the fibers loop around the lug, transferring load without an adhesive interface. The Kronos approach for coil support likely involves **preloading**: they mention "cryogenic pistons" that set the preload on structure[100][101]. Those pistons could be screws that push on a composite ring which in turn presses on the coil – in assembly, one might cool the piston with dry ice (hence "ice-powered") to shrink it, tighten, then let it warm and expand to lock in stress[102].

Specialty processes like **spark plasma sintering (SPS)** or **field-assisted sintering** are being explored to join materials quickly by pulsed current. One study used FAST (field-assisted sintering technique) to diffusion bond tungsten to EUROFER steel at lower temperatures by applying a current and pressure, achieving good bonds with a Cu interlayer[103][104]. Such techniques might allow attaching small robust HfC tiles to a Cu heat sink in a divertor finger with less thermal damage.

Additive meets assembly: Interestingly, additive manufacturing can also help in multi-material integration. One can directly 3D-print graded materials – e.g. a DED machine could feed gradually from HEA powder to ceramic powder to create a gradient. Or one can print complex shapes that act as interlocking interfaces (like dovetails or lattice textures) to mechanically key materials together.

Quality control and scalability remain challenges. Diffusion bonds require extremely clean surfaces and precise temperature control to avoid voids or embrittlement[105][106]. Additive parts must be inspected for porosity and residual stress. Thick ceramic coatings need thorough adhesion testing (e.g. thermal cycling to see if they spall). In a fusion reactor, any joint or interface becomes a potential failure point under thermal cycling and irradiation. Thus, current R&D is focusing on **joining under fusion-relevant conditions** – e.g., how does a W–steel diffusion bond hold up after 100 dpa and thousands of thermal cycles? Preliminary results show that with proper interlayers (like a ductile V or Cu layer), joints can remain intact where direct bonding would crack[107][106].

In summary, advanced fabrication techniques such as **MOCVD, sputtering, additive manufacturing (SLM, DED, binder jet), and diffusion bonding** are crucial for harnessing the novel materials in fusion. They allow engineers to build complex, multi-material assemblies like a first wall panel that has a HfC–N protective coat on a CrMoNbV lattice structure, or a TF coil that combines ReBCO tape, graphene-Al stabilizer, MP35N reinforcement, all encapsulated by a carbon fiber shell. Integration is as important as the material properties themselves – a superb material poorly joined can fail early. The ongoing progress in these manufacturing technologies gives confidence that by the time aneutronic tokamaks are ready for construction, the means to fabricate them will also be ready.

Synergistic Design and Multi-Material Assemblies

The true strength of these advanced materials emerges when they are **combined in an integrated design**, each contributing to an aspect of reactor performance. Aneutronic tokamaks like the Kronos S.M.A.R.T. concept explicitly adopt a *multi-material architecture*: a mosaic of ceramics, metals, and composites, each placed where they are most effective, working in concert under the harsh fusion environment. This section explores how the previously discussed materials synergize in a reactor system – how their properties complement each other to meet the simultaneous demands of high temperature, high stress, and high particle flux. We also highlight specific assembly concepts (from Kronos and others) that exemplify this synergy, such as the layered **outer wall/DEC module** and the "barrel and hoop" coil support analogy[88][13].

First, consider the **plasma-facing first wall and direct energy converter (DEC)** modules. In an aneutronic reactor, the first wall is not just a sacrificial shield as in ITER; it is an active component of power

conversion, directly exposed to MeV alphas that carry >95% of the fusion energy[14][15]. Kronos envisions a multi-layer first wall where HfC–N or Ta–Hf–C ultra-high-temperature ceramic forms the inner surface (facing the plasma), backed by a high-conductivity structure, and integrated with electrostatic/magnetohydrodynamic conversion elements[108][74]. The **HfC–N layer** provides the necessary resilience: it handles the 10^8 K plasma exposure and repetitive thermal shocks without melting or eroding rapidly[1][26]. Its sputter resistance ensures low impurity influx to the plasma and minimal surface recession over time[27]. Behind it, a **structural layer of CrMoNbV HEA** or similar supports the ceramic. The HEA's job is to maintain mechanical integrity – since it suffers negligible neutron embrittlement or helium swelling[25], it can be a lifetime component. It acts as a ductile, tough "backbone" that the ceramic is bonded to, absorbing any forces (e.g. disruption-induced impulses or handling loads). The HEA's high thermal conductivity (CrMoNbV is primarily metallic, so likely ~20 W/m·K at 700 °C) and high allowable temperature (≥800 °C) mean it can serve as a **heat spreader and primary heat sink**. Heat that conducts through the ceramic flows into the HEA structure, which is cooled by liquid metal or high-temperature gas cooling loops. Because the HEA can run hot, the coolant can be at ~700 °C (Brayton cycle friendly) – a big efficiency win over water cooling at ~150 °C in ITER.

Next, integrated into this wall are **direct energy conversion elements**. For example, an electrostatic DEC might involve metal foils or grids at different potentials to decelerate and collect alpha particles. Those foils could themselves be made of advanced materials: perhaps a fine mesh of Mo or Cu laminated with a monolayer of graphene to reduce secondary electron emission and withstand heating. Or an MHD channel could be embedded, where the flowing plasma (or a generated conducting fluid) interacts with a ~30 T magnetic field to generate electricity[109][110]. In that case, the **MHD electrodes and channel walls** would need the ultra-refractory ceramic coatings (HfC to survive plasma contact) and the structural HEA or composite to support them. Kronos documentation hints at such complexity: the outer wall is described as "not a passive boundary – it is a multi-layered engine of confinement and conversion"[111][112], indicating layers including graphene-based diagnostics, thermionic converters, and bremsstrahlung absorbers. Graphene appears even here: a *graphene diagnostic network* in the wall can quickly sense plasma thermal radiation or field changes[73][113]. Graphene's radiation tolerance is high (being carbon) and it can form transparent conductive films; thus, it could serve as distributed sensors or electrodes that survive neutron/gamma flux.

Now, moving to the **magnet system and support structure**, synergy is evident in the **"bucking cylinder + outer shell"** construct[114][76]. The internal ceramic or CFRP bucking cylinder (potentially alumina or carbon fiber composite) endures the pure compression of inward coil forces. Ceramic is ideal in compression (alumina's compressive strength is ~2 GPa) and won't creep at cryogenic conditions. Carbon fiber (if used) adds toughness and nearly zero thermal strain, ensuring the cylinder remains tight from room temp to 4 K. The external MP35N or steel compression shell, possibly overwrapped with carbon fiber, contains the outward forces. By having these two elements, the coils themselves (made of brittle ReBCO superconductors embedded in a soft copper matrix) are kept in a compressive preload and isolated from tension[85][13]. This is crucial because HTS tapes can handle compression better than tension. The synergy here: **MP35N** provides huge tensile strength to the outer shell (like a steel band around a barrel), **CFRP** provides stiffness and thermal fit, and even **HEA or steel flanges** might integrate to transfer loads between modules. Kronos specifically notes that using both bucking cylinder and shell "significantly increase mechanical margin and ensure no part of the coil experiences strain beyond 0.4%"[13]. Without both, you'd overstress something; together they share load optimally.

Looking at **thermal management**, consider a quench event in a coil. The combination of **graphene–Al conductors and MP35N structure** handles it gracefully. When a section quenches, current diverts into the graphene–Al stabilizer which has low resistance and high thermal conductivity, quickly spreading the heat and current. The MP35N band, being metallic, can also carry some current if needed and will absorb elastic

energy as the coil tries to expand thermally (MP35N's high yield strength means it can momentarily take extra stress, acting like a buffer spring). Meanwhile, carbon fiber composite components, with low thermal expansion, will not suddenly contract or expand, reducing mechanical shock to the system. Thus, each part contributes: graphene–Al keeps the coil electrically safe, MP35N keeps it mechanically safe, CFRP keeps it geometrically stable, and HEA parts (if any in the support) remain unfazed by any neutron burst or temperature excursion within design.

Another synergy is in **radiation and activation management**. By using low activation materials (C, V, Cr, etc.), the entire reactor can potentially be designed for easier end-of-life handling. For example, instead of a huge steel vacuum vessel that becomes intermediate-level waste, a CrMoNbV HEA vessel will have little long-term radioactivity[44]. Carbon fibers and ceramics like HfC produce essentially no activation. So the only significant activated materials would be maybe some structural steel or Cu. This means the multi-material approach isn't just about performance, it's about **sustainability of fusion** – enabling quicker maintenance turnaround and decommissioning. If a first wall module is made of HEA + HfC, after a campaign it might be reprocessed rather than disposed, since the constituents are (by design) not forming long-lived isotopes.

One illustrative synergy from Kronos: they mention their first wall could last >5 years between overhauls with HfC–N, vs ITER's W divertor needing replacement ~every 1–2 years[12][115]. This extension comes from combining high-temp capability (less thermal stress per pulse), sputter resistance (slower erosion per pulse), and a supportive structure (less cracking/fatigue). They also suggest the HEA vessel might last the full plant lifetime with negligible embrittlement[12]. If realized, that's a revolutionary improvement – it takes the material replacement schedule off the critical path of plant operation.

Integration into ReBCO magnets and DEC modules demands thinking in terms of **assemblies** instead of individual materials. For example, a DEC module might be a sandwich: HfC–N layer | HEA substrate | graphene-Al electrode | insulation | composite frame. Each layer must be compatible with neighbors: thermal expansion, chemical, electrical compatibility. This is where intermediate materials and surface treatments come in. You might coat the HEA with a thin diffusion barrier (e.g. TiN or SiC) before putting HfC on it to prevent carbide formation at the interface that could weaken it. Or when casting HEA around a composite insert, you cool the process to not char the composite.

Case study: Kronos SMART outer wall – It reportedly incorporates a graphene-based diagnostic layer for sensing and perhaps current collection[111][112]. Graphene is transparent to most radiation and thin, so embedding a network of graphene sensors in the first wall could allow real-time mapping of heat flux (via changes in resistance) or magnetic fluctuations. These sensors would be printed onto an insulating substrate (maybe Al_2O_3 or polymer) which is bonded on the cooler side of the HEA wall. This is a good example of layering: the hot face has HfC for protection, the middle is HEA for structure, and the back has graphene sensors and maybe antennae for communications – a **smart wall** concept.

All these synergistic designs need **systems engineering**: you must analyze combined loads – thermal, mechanical, and electromagnetic – as one. Finite element models of the entire coil-plasma-vessel assembly are used to ensure that when, say, the plasma heats the wall by 100 °C and the magnets pulse, the differential expansions and contractions do not misalign the coils or crack the coatings. Multi-material assemblies often incorporate **flexibility in specific directions**: e.g., sliding interfaces or flex joints to accommodate small mismatches. A practical note: where flexibility is needed, one might intentionally use a lower-stiffness material (maybe a small layer of fiberglass or a thin Inconel flexure) to decouple materials. That way the stiff materials (ceramics, CF) aren't overstressed by being too rigidly bound to metals.

In conclusion, the synergy of advanced materials in fusion is akin to an **orchestra**: each material plays a part in harmony. The ceramics handle the brunt of plasma heat and erosion, the HEAs and superalloys

handle structural loads and radiation, the graphene and composite elements manage electrical/thermal stability and precise support, and together they enable a reactor to operate in a regime none of them could achieve alone. The result, as Kronos and others predict, is a **higher-performance fusion system**: one that can run hotter (first wall at 1000 °C vs 300 °C) and at higher field (20–30 T vs ~12 T)[116][117], delivering more power from a smaller device. Achieving this requires deep integration of materials science and reactor engineering – a "materials-first design philosophy" that Kronos explicitly notes as justified by these breakthroughs[118][119]. The materials revolution thus paves the way for aneutronic fusion to leap from theoretical possibility to practical reality.

Research Gaps and Future Directions

While the prospects of HfC ceramics, HEAs, graphene conductors, and carbon composites in fusion are exciting, significant **research gaps** remain. Bridging these gaps will be essential to qualify these materials for use in a commercial reactor and to discover even better alternatives. We outline the key areas for future work:

1. Long-duration irradiation testing: Many claims about HEAs and UHTCs in fusion rely on short-term ion irradiations or modeling. We need neutron source testing (e.g. in a facility like IFMIF or a multi-MW fission reactor) to expose candidate materials to 10s of dpa and high helium appm levels. *Does HfC–N retain its properties after 5 years of mixed D–^{3}He and neutron bombardment?* Will it undergo radiation-induced amorphization or high-temperature helium embrittlement? Some carbides can swell or graphitize under extreme neutron dose. Similarly, *can CrMoNbV HEA maintain its phase stability up to 100 dpa and 600 °C?* Although theory suggests entropy stabilizes it[42], transmutation could introduce impurities that precipitate. Dedicated campaigns to irradiate miniaturized samples of these materials, perhaps in surrogate spectra (proton or heavy ion for surrogate damage plus helium injection), will provide data on swelling, hardness changes, phase stability, and thermal properties post-irradiation[120][121]. There is also essentially no data on *graphene or carbon composites under neutron irradiation*. Graphite is known to withstand high doses (with some property swings), but resin matrices in CFRP will degrade with fast neutrons and gamma – new radiation-resistant polymers or ceramic matrices might be needed if composites are to be in high flux zones.

2. Plasma-material interaction studies with advanced fuels: Aneutronic plasmas have different spectra (more alphas, perhaps higher electron temperatures because less dilution). We need experiments in *linear plasma devices or tokamak test stands* to expose HfC–N and Ta–Hf–C samples to D–He or p–B plasmas. The questions to answer: *What is the sputtering yield with 3.6 MeV alphas hitting HfC?* (Probably low, but needs measurement). *Does the carbonitrided surface form any unwanted coatings (like Boron from p–^{11}B reactions) or undergo chemical reactions with helium?* Tritium retention is less an issue for aneutronic, but helium implantation could cause fuzz or blistering on carbide surfaces at certain temperatures; understanding that will guide temperature windows for operation. Additionally, if direct energy conversion uses electrodes in plasma, their work function and secondary electron emission in a fusion environment need study – e.g. does HfC's 3.4 eV work function hold when the surface is saturated with helium or boron? These can be investigated with in-situ diagnostic experiments (work function probes, laser thermionic emission tests) on exposed surfaces.

3. Component-scale demonstrations: It's one thing to test coupons; it's another to fabricate a realistic component and see it perform. Intermediate-scale projects, like **plasma-facing component mockups**, **magnet sub-assemblies**, or **vacuum chamber sectors**, should be built using these materials to uncover practical issues. For example, a small high-field test magnet could be wound using ReBCO with graphene-Al stabilizer and MP35N overbanding, then ramped to failure to see if the design predictions hold (did the strain stay under 0.4% everywhere? Did the quench detect and spread properly?). Or a "mini first wall"

module with HfC–HEA layered structure could be put in a plasma gun facility to endure cyclic thermal loads. These prototypes will help refine manufacturing and joining approaches under realistic conditions. Kronos is reportedly advancing materials through coupon irradiation and plasma testing[55] – expanding that to integrated assembly tests will build confidence.

4. Multi-material joining and thermal cycling: As discussed, diffusion bonding and graded joints are being developed to join materials like W and steel[94][98]. Similar R&D must be done for HfC/HEA, HEA/steel, composite/metal joints that will be present. Extensive **thermal cycling tests** (e.g. cycling a bonded assembly from room temp to 1000 °C repeatedly, or from 300 K to 4 K and back many times) are needed to ensure joints don't fatigue. The interplay of differential expansion, irradiation, and creep at joints is complicated – modeling (with tools like finite element plus phase field methods) combined with experiments will be needed. For instance, how does an HfC coating on HEA behave after 1000 cycles of rapid heating to 1700 °C and cooling? Any spallation or delamination? If so, interlayers or surface texturing might be required.

5. Scalability and manufacturability: Some materials are currently lab curiosities (e.g. gram-scale HEA buttons, or 2-inch CVD HfC coatings). We need to scale these to tonnage and square-meter scales respectively. This involves materials processing research: can we produce HEA powders cheaply (perhaps by carbothermic reduction of mixed oxides, then alloying) for use in additive manufacturing? Can we develop **spray coatings or paint-on slurries** for HfC that can cover large areas and then be laser-fused in place? Also, **cost reduction** is a big factor. Using exotic elements like hafnium or tantalum could be very costly at reactor scale; future research should explore if more abundant alternatives (zirconium carbide? tungsten alloyed with small %HfC?) can approximate the performance at lower cost, or if recyclability can mitigate cost (e.g. recovering and reusing Ta–HfC tiles).

6. AI-driven materials discovery: Given the vast combinatorial space of multi-component alloys and ceramics, **machine learning and high-throughput computation** can expedite finding optimal compositions. As noted, HfC–N was a fairly recent discovery as a record melting compound[122]. There could be other mixtures – perhaps adding a bit of titanium or boron to HfC–N might improve its toughness or reduce oxidation. Similarly, HEAs permit endless tuning: adding 1–2% of an element like Ti or Hf might improve grain boundary cohesion or radiation recombination. *Computational thermodynamics (CALPHAD) plus DFT defect calculations* can screen thousands of compositions for desired traits (like low vacancy formation energy, implying better self-healing under radiation[123]). Several research groups are indeed using AI to identify "radiation super-alloys"[122] and to design materials with targeted thermal conductivity or thermal expansion. Fusion programs should invest in such **data-driven materials R&D**, as the timeline for commercial fusion can be shortened if we rapidly iterate to find materials that meet all criteria rather than testing one-by-one for decades.

7. In-situ health monitoring and adaptive control: Lastly, future directions include embedding sensors and developing **self-healing or self-adaptive materials**. Kronos mentions V_2O_3 (vanadium sesquioxide) as an "adaptive" material in coils[54] – V_2O_3 has a metal-insulator transition around 160 K, so maybe they envision it as a smart resistor that changes with temperature to aid quench protection. This is an example of functional materials being integrated. Research could explore, say, **self-healing coatings** that release a sacrificial layer upon overheating (like a thermal paint that evaporates to protect the base). Or **optical fiber sensors** woven into composites to detect strain and radiation in real time. In-situ monitoring (AI interpreting fiber Bragg gratings signals, etc.)[73][124] can give early warning of material degradation, enabling predictive maintenance. Developing materials that not only withstand extremes but also *indicate* their status is a promising field.

In conclusion, while the materials reviewed are at the cutting edge, moving from promising samples to a working fusion core requires bridging scientific and engineering gaps. Priorities should be: obtaining **irradiation data** to validate longevity, perfecting **joining and assembly** methods for multi-material structures, and leveraging **modern computational tools** to optimize compositions and detect failure precursors. International collaborations, such as materials test reactors and simulation hubs, will be crucial – e.g., the planned fusion prototyping facilities could incorporate testing of these advanced materials in their mission. The **transformative potential** of these materials justifies a concerted research effort: they address what used to be "showstoppers" for fusion (first wall erosion, magnet limits, etc.), so filling the knowledge gaps is an essential step on the road to a viable aneutronic fusion reactor.

Risk Acknowledgment and Ongoing Research Disclosure

The Kronos team is acutely aware that the S.M.A.R.T. reactor's novel architecture entails significant technical risks. Every core design choice – from the 30+ tesla REBCO magnet system to the aneutronic D–^{3}He/p–^{11}B fuel cycle and integrated direct energy conversion – pushes beyond prior fusion engineering experience. Key risk factors include the structural and quench limits of brittle high-field superconductors, the untested long-term plasma stability of an ultra-compact, high-β *aneutronic* core, and the durability of first-of-a-kind plasma-facing direct conversion systems. **Proprietary risk-mitigation programs** are in place targeting materials, controls, diagnostics, and field validation. For example, non-insulated HTS magnet windings and novel support structures have been adopted to reduce quench risk and mechanical fatigue in the 30 T toroidal field coils[1]. Concurrently, advanced **graphene-coated diagnostics** and AI-driven control algorithms are being developed to manage plasma instabilities in real-time, ensuring that any onset of anomalous behavior is detected and corrected within microseconds[2]. Each subsystem – magnets, plasma, and conversion modules – is first proven in high-fidelity **digital twin simulations** and then in scaled physical prototypes to validate performance and safety margins. This simulation-first, AI-assisted development pipeline has already compressed prototyping timelines by >75%[3]. Historical precedents guide our approach: just as high-field tokamak projects like MIT's SPARC have worked to retire magnet and confinement risks through aggressive testing and modeling[4][5], Kronos is methodically probing the limits of our design in silico and in the lab before any full-scale build. We also draw lessons from past direct-conversion experiments (e.g., the LLNL "Venetian blind" electrostatic collector on TMX achieved ~50–65% efficiency in the 1980s[6]) and advanced reactor studies (the Apollo-L3 D–^{3}He concept projected ~80% photon-to-electric conversion[7]). These analogs underscore both the potential payoffs and the technical challenges, informing our risk models. **Roadmap and Disclosure:** All critical risks are tracked against quantitative resolution milestones on our 2025–2032 technology roadmap. We have instituted a rigorous test campaign for each milestone – from subcoil stress tests and plasma module mockups to integrated system trials – with clear exit criteria before scaling up. By 2030, the team will publish detailed performance data and **risk-retirement metrics** for each innovation, demonstrating which risks have been resolved and which remain in progress. In parallel, independent experts will be engaged in design reviews to ensure no failure mode is overlooked. In summary, Kronos Fusion Energy acknowledges that our architecture's bold leaps come with considerable risk; we have architected a comprehensive, AI-supported risk mitigation strategy to match. **Validated results** from our digital twins and prototype subsystems will be transparently reported by 2030, well before commercialization, and will include complete data on materials endurance, plasma stability, conversion efficiency, and system control reliability. This ongoing research and risk disclosure process reflects our commitment to rigorous engineering practice and to earning stakeholder confidence through evidence-based progress.

[1] C. F. Weggel et al., The Kronos Fusion Architecture: A New Standard for Aneutronic Energy Systems, Kronos Fusion Energy, whitepaper, 2025.

[2] W. L. Barr and R. W. Moir, "Test results on plasma direct conversion," Journal of Fusion Energy, vol. 2, pp. 131–139, 1982. *https://link.springer.com/article/10.1007/BF01314197*
[3] G. L. Kulcinski, J. F. Santarius, and L. M. Waganer, "Apollo-L3 – An advanced fuel fusion reactor utilizing direct and thermal energy conversion," Fusion Technology, vol. 19, no. 4, pp. 791–797, 1991. *https://www.tandfonline.com/doi/abs/10.13182/FST91-A29441*
[4] A. J. Creely et al., "Overview of the SPARC tokamak," Journal of Plasma Physics, vol. 86, no. 5, 865860502, 2020. *https://doi.org/10.1017/S0022377820001257*
[5] S. J. Zinkle and N. M. Ghoniem, "Operating temperature windows for structural materials in fusion reactors," Fusion Engineering and Design, vol. 51–52, pp. 55–71, 2000. Available: *https://www.sciencedirect.com/science/article/abs/pii/S0920379600001625*
[6] R. D. Moir, J. Hovingh, T. K. Fowler, D. E. Baldwin, et al., "Venetian blind direct converter experiments on the TMX-U tandem mirror," Fusion Engineering and Design, vol. 8, pp. 25–33, 1989. *https://www.sciencedirect.com/science/article/abs/pii/0920379689900577*
[7] H. Matsumoto, K. Yamada, K. Nishikawa, et al., "Recent progress of thermionic energy conversion for fusion reactors," Frontiers in Mechanical Engineering, vol. 3, article 13, 2017. *https://www.frontiersin.org/articles/10.3389/fmech.2017.00013/full*

Conclusion

Advanced materials are redefining what is possible in fusion reactor design. In this paper, we have shown that the transition from conventional D–T tokamaks to **compact aneutronic tokamaks** is fundamentally enabled by a suite of new materials: ultra-refractory carbides (HfC-based), high-entropy alloys (CrMoNbV class), cryogenic metal composites (MP35N and graphene-enhanced aluminum), and high-modulus carbon fiber composites. Individually, each of these materials offers a leap in performance over the status quo – whether it be a ~1000 K increase in allowable wall temperature with HfC–N vs tungsten[63][1], an order-of-magnitude reduction in radiation-induced swelling with HEAs vs steels[4][25], a doubling of stabilizer thermal conductivity with graphene vs pure aluminum[8], or near-zero thermal distortion of coil supports with carbon fibers vs metallic supports[11]. More importantly, **in combination, these materials unlock a new regime of reactor operation**: one with plasma-facing components that can handle steady 1000–1700 °C surfaces, structural shells that remain strong after years of 14 MeV proton irradiation, magnets that reach 30 T without self-destructing, and structural frameworks that maintain alignment and integrity through thermal cycles and pulsed loads.

By integrating these materials into designs like the Kronos S.M.A.R.T. architecture, engineers are able to push key performance metrics: higher field (hence higher plasma pressure for a given size), higher temperature operation (for direct conversion and efficiency), and longer component lifetimes[12][117]. For example, using HfC–N coatings on the first wall and divertor, a reactor can run a hotter edge plasma without melting the wall, allowing advanced exhaust schemes and higher core purity (since less sputtered impurity)[27][26]. Employing CrMoNbV HEA for the vacuum vessel and internals means these parts can be designed thinner (due to higher strength at temperature) and won't need frequent replacement – reducing downtime and waste[12]. The combination of MP35N and graphene-Al in the magnet system permits HTS coils to achieve their full potential in field and current density, confident that mechanical and quench stability limits are addressed. And carbon fiber composites in support structures ensure that precise geometries (coil spacing, alignment of direct converters, etc.) are maintained from assembly, through cooldown, and throughout operation, providing a stable platform for the plasma and equipment.

In summary, this **materials revolution** is turning what were once regarded as insurmountable problems for aneutronic fusion – plasma-facing erosion, neutron damage, magnet stress – into tractable engineering

challenges. Each material acts as a "technology enabler": HfC–N enables >95% direct energy recovery without solid wall meltdown[15][125]; HEAs enable a structural lifetime matching the plant lifetime with minimal maintenance[12]; advanced conductors enable magnet ramps and shots at performance that would fry traditional coils. The result is a fusion plant paradigm very different from the massive, low-power-density ITER-like machines. We are looking at smaller, more efficient reactors that can realistically approach **commercial viability** in markets from terrestrial power to space propulsion, precisely because these materials remove some of the toughest constraints (like the low duty cycle of replacing inner components, or the giant radial build needed for shielding fragile materials).

That said, realizing this vision demands continued interdisciplinary effort. We highlighted research gaps such as radiation testing, fabrication scaling, and materials discovery. Addressing these will require coordinated programs: irradiation facilities to qualify new materials under fusion conditions, development of robust joining techniques to assemble multi-material components, and leveraging AI/quantum computing for materials modeling. We also stress the importance of **system-level testing** – integrated demos that combine these materials in prototypical environments to catch any unforeseen interactions.

The pathway to **commercial aneutronic fusion** will be accelerated by prioritizing materials R&D as highly as plasma physics. The payoff is immense: if we can demonstrate that these advanced materials hold up, the remaining physics challenges (like steady-state D–^{3}He plasma sustainment) can be tackled on a much shorter timeline than the half-century saga of adapting 20th-century materials to fusion. With HfC, HEAs, graphene, and carbon composites in our toolkit, we have, for the first time, a materials basis that *matches the extreme nature of fusion energy*. As we continue to innovate, it is conceivable that even more revolutionary materials (quantum composites, self-healing structures) will come into play, guided by the success of the current quartet.

In conclusion, the integration of these advanced materials is **transformative** – it shifts fusion engineering from the paradigm of "how do we protect our materials from the fusion plasma" to "how do we design materials that harness the fusion plasma". This shift will be a decisive factor in making aneutronic, high-efficiency fusion not only scientifically intriguing but *commercially achievable*. The strategic priority moving forward is clear: invest in the materials that make fusion work. The reactor of the future will not be built of steel and copper, but of HfC and HEAs, of graphene and carbon fiber – a high-performance symphony of the elements enabling the power of the stars here on Earth.

BIBLIOGRAPHY

Materialia, Fusion Engineering and Design, Journal of Nuclear Materials, etc., as cited in the text above.)[1][3][4][6][8][9]

[3] [45] Computational discovery of ultra-strong, stable, and lightweight refractory multi-principal element alloys. Part II: comprehensive ternary design and validation | npj Computational Materials

https://www.nature.com/articles/s41524-023-01031-6?error=cookies_not_supported&code=da13d61a-901a-4f30-ac84-f452fabe1257

[4] [5] [51] [52] [120] pure.hud.ac.uk

https://pure.hud.ac.uk/files/35489210/Paper_HEA_He_submitted.pdf

[6] (PDF) Mechanical properties of MP35N as a reinforcement material ...

https://www.researchgate.net/publication/224055583_Mechanical_properties_of_MP35N_as_a_reinforcement_material_for_pulsed_magnets

[7] Cryogenic mechanical properties of superalloy MP35N - ScienceDirect

https://www.sciencedirect.com/science/article/abs/pii/0011227584900407

[8] [65] [68] [69] [70] [71] chalcogen.ro

https://chalcogen.ro/87_VellliangiriM.pdf

[9] [75] Copper-graphene composites could lead to better electrical wires and motors | Graphene-Info

https://www.graphene-info.com/copper-graphene-composites-could-lead-better-electrical-wires-and-motors

[10] [11] Présentation PowerPoint

https://toray-cfe.com/wp-content/uploads/2025/02/Torayca-M46J-Technical-Data-Sheet.pdf

[18] [19] [20] [42] [43] [44] [48] High-entropy alloys for nuclear applications | Journal of Materials Science

https://link.springer.com/article/10.1007/s10853-024-10511-z

[21] [29] Hafnium Carbide - Shanghai Greenearth Chemicals Co.,Ltd

https://www.greenearthchem.com/product/hafnium-carbide

[23] High Purity Tantalum Hafnium Carbide Powder

https://princetonpowder.com/tantalum-hafnium-carbide-powder/

[24] Tantalum hafnium carbide - wikidoc

https://www.wikidoc.org/index.php/Tantalum_hafnium_carbide

[30] [31] [32] [35] [36] A comprehensive screening of plasma-facing materials for nuclear fusion

https://arxiv.org/html/2407.00858v1

[34] Ultra-high-temperature tantalum-hafnium carbonitride ceramics ...

https://www.sciencedirect.com/science/article/abs/pii/S0272884221022227

[37] Thermal properties of Cf/HfC and Cf/HfC-SiC composites prepared ...

https://www.sciencedirect.com/science/article/abs/pii/S0955221917308580

[38] [41] (PDF) Electron emission from hafnium carbide

https://www.researchgate.net/publication/251942010_Electron_emission_from_hafnium_carbide

[39] Electron emission from HfC(100) single-crystal tip - AIP Publishing

https://pubs.aip.org/avs/jvb/article/42/4/043204/3303195/Electron-emission-from-HfC-100-single-crystal-tip

[40] Use of HfC(310) as a High Brightness Electron Sources for ...

https://academic.oup.com/mam/article-pdf/20/S3/48/48284692/mam0048.pdf

[46] Probing phase stability in CrMoNbV using cluster expansion method ...

https://www.researchgate.net/publication/371432428_Probing_phase_stability_in_CrMoNbV_using_cluster_expansion_method_calphad_calculations_and_experiments

[47] A comprehensive review on the deformation behavior of refractory ...

https://mfr.edp-open.org/articles/mfreview/pdf/2023/01/mfreview220083.pdf

[49] High-Entropy Alloys for Advanced Nuclear Applications - PMC

https://pmc.ncbi.nlm.nih.gov/articles/PMC7827623/

[50] Diffusion controlled helium bubble formation resistance of FeCoNiCr ...

https://www.researchgate.net/publication/335064298_Diffusion_controlled_helium_bubble_formation_resistance_of_FeCoNiCr_high-entropy_alloy_in_the_half-melting_temperature_regime

[53] Radiation Resistance of High-Entropy Alloys CoCrFeNi and ... - MDPI

https://www.mdpi.com/1996-1944/17/19/4751

[56] [57] [93] Additive Manufacturing Technologies of High Entropy Alloys (HEA): Review and Prospects - PMC

https://pmc.ncbi.nlm.nih.gov/articles/PMC10057660/

[58] About this Abstract - ProgramMaster.org

https://www.programmaster.org/PM/PM.nsf/ApprovedAbstracts/038019CC8709E04485258B430064707C?OpenDocument

[59] [PDF] Design considerations for high entropy alloys in advanced nuclear ...

https://www.osti.gov/servlets/purl/1981686

[60] [PDF] Mechanical Properties of a Ni-Co-Cr-Mo Alloy

https://public.magnet.fsu.edu/AToth/NSF%20publication%20access/2025/05%20Newly%20Entered%20Pubs%20May%202025/2025%20Cited/2025_Mechanical%20Properties.pdf

[64] [1704.04275] Low temperature physical properties of Co-35Ni-20Mo-10Cr alloy MP35N

https://arxiv.org/abs/1704.04275

[66] [72] Thermal conductivity of Aluminum/Graphene metal-matrix composites

https://www.sciencedirect.com/science/article/abs/pii/S2352492822000253

[67] Graphene-Aluminum Composites: Supercharging Conductivity

https://elkamehr.com/en/graphene-aluminum-composites-supercharging-conductivity/

[77] [78] Toray Launches Carbon Fiber with High Tensile Modulus and Enhanced Strength

https://www.specialchem.com/polymer-additives/news/toray-carbon-fiber-high-tensile-modulus-000232909

[79] [PDF] Transverse Properties of Carbon Fibres by Nano-Indentation and ...

https://archimer.ifremer.fr/doc/00000/4502/4013.pdf

[82] Paper Fatigue properties of unidirectional carbon fibre composites at ...

https://www.sciencedirect.com/science/article/abs/pii/001122759190086C

[83] [PDF] Properties of cryogenic and low temperature composite materials

https://researchportal.bath.ac.uk/files/217611109/Properties_of_cryogenic_and_low_temperature_composite_materials_A_review.docx_1_.pdf

[89] Cryogenic Impact on Carbon Fiber-Reinforced Epoxy Composites ...

https://www.mdpi.com/2504-477X/8/11/459

[90] Additive manufacturing of high-entropy alloys: Current status and ...

https://www.sciencedirect.com/science/article/pii/S2772810224000151

[91] Additive Manufacturing of High-Entropy Alloys: Microstructural ...

https://link.springer.com/article/10.1007/s11669-021-00913-w

[92] Strong yet ductile nanolamellar high-entropy alloys by additive ...

https://www.nature.com/articles/s41586-022-04914-8

[94] [95] [96] [97] [98] [99] [105] [106] [107] Evaluation of Tungsten—Steel Solid-State Bonding: Options and the Role of CALPHAD to Screen Diffusion Bonding Interlayers

https://www.mdpi.com/2075-4701/13/8/1438

[103] Exploring FAST Technique for Diffusion Bonding of Tungsten to ...

https://pmc.ncbi.nlm.nih.gov/articles/PMC11174111/

[104] Evaluation of Tungsten—Steel Solid-State Bonding

https://www.ornl.gov/publication/evaluation-tungsten-steel-solid-state-bonding-options-and-role-calphad-screen-diffusion

[121] Compositional Effects of Additively Manufactured Refractory ... - OSTI

https://www.osti.gov/pages/biblio/1872238

[123] Development of reduced-activation and radiation-resistant high ...

https://www.researchgate.net/publication/379523917_Development_of_reduced-activation_and_radiation-resistant_high-entropy_alloys_for_fusion_reactor

TABLES

Table 1: Advanced materials and validated properties for reactor application. The table lists high-performance materials—including $HfC_{0.75}N_{0.22}$, Ta_4HfC, CrMoNbV HEA, MP35N, V_2O_3, graphene–Al composites, and carbon fibers—along with their key properties, reactor applications, and roles in supporting >30 T fields and >1000 °C operation. 4

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The Materials Revolution in Fusion

HfC, CrMoNbV High-Entropy Alloys, Graphene-Aluminum, and Structural Carbon Composites

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Detailed Technical Appendix Available Upon Request
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